

Representative Structural Calculation Package

Long-span warehouse

Item	Value
Project	Long-span warehouse
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	05-long-span-warehouse

Nodes	Members	Loads	Combos	Max disp.	Max util.
30	37	104	28	12.77 mm	0.539

Model And Design Basis

Item	Value
Building type	industrial
Description	Single-story long-span industrial warehouse with wide transverse bays.
Occupancy	Warehouse
Bounds X/Y/Z	32.00 / 24.00 / 8.000
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	22	0.000 kN	0.000 kN	-2304 kN
L	live	22	0.000 kN	0.000 kN	-614.40 kN
WX	wind	15	192.00 kN	0.000 kN	0.000 kN
WY	wind	15	0.000 kN	199.11 kN	0.000 kN
EX	seismic	15	149.91 kN	0.000 kN	0.000 kN
EY	seismic	15	0.000 kN	149.91 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	768.00	2304 kN	614.40 kN	192.00 kN	199.11 kN

Load Derivation Formula Trace

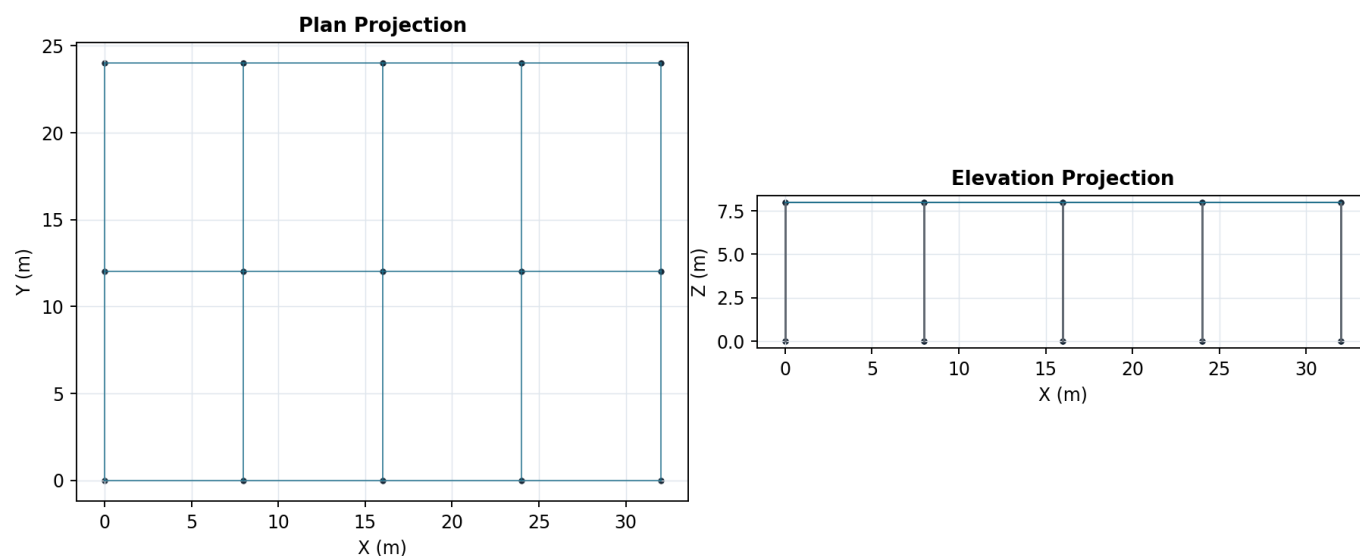
ID	Case	Formula	Result	Inputs
basis-occupancy	-	occupancy preset lookup	-	occupancy=warehouse
basis-plan-area	-	$\max(\text{model width} * \text{model depth}, 1)$	768.00 m2	Bx=32.00 m, By=24.00 m
basis-roof-area	-	roofArea input or typical floor area	768.00 m2	roofArea=768.00 m2
D-ST1	D	$A * q_D$	2304 kN	A=768.00 m2, $q_D=3.000 \text{ kN/m}^2$
L-ST1	L	$A * q_L$	614.40 kN	A=768.00 m2, $q_L=0.800 \text{ kN/m}^2$
WX-ST1	WX	$p_{WX} * B_y * h$	192.00 kN	$p_{WX}=1.000 \text{ kN/m}^2$, $B_y=24.00 \text{ m}$, $h=8.000 \text{ m}$
WY-ST1	WY	$p_{WY} * B_x * h$	199.11 kN	$p_{WY}=0.778 \text{ kN/m}^2$, $B_x=32.00 \text{ m}$, $h=8.000 \text{ m}$
EX-BASE	EX	$CsX * \sum(W_i)$	149.91 kN	$CsX=0.061 \text{ g}$, $\sum W=2458 \text{ kN}$
EY-BASE	EY	$CsY * \sum(W_i)$	149.91 kN	$CsY=0.061 \text{ g}$, $\sum W=2458 \text{ kN}$
EX-ST1	EX	$V_x * W_i * z_i / \sum(W_i * z_i)$	149.91 kN	$V_x=149.91 \text{ kN}$, $W_i=2458 \text{ kN}$, $z_i=8.000 \text{ m}$
EY-ST1	EY	$V_y * W_i * z_i / \sum(W_i * z_i)$	149.91 kN	$V_y=149.91 \text{ kN}$, $W_i=2458 \text{ kN}$, $z_i=8.000 \text{ m}$

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

Long-span warehouse



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	9.591 mm	0.455	0.000
KDS-ST-02	OK	11.14 mm	0.529	0.000
KDS-ST-04-WX-P	OK	11.59 mm	0.477	0.000
KDS-ST-04-WX-N	OK	11.59 mm	0.477	0.000
KDS-ST-04-WY-P	OK	12.77 mm	0.539	0.000
KDS-ST-04-WY-N	OK	12.77 mm	0.539	0.000
KDS-ST-05-EX-P	OK	11.01 mm	0.477	0.000
KDS-ST-05-EX-N	OK	11.01 mm	0.477	0.000
KDS-ST-05-EY-P	OK	11.78 mm	0.524	0.000
KDS-ST-05-EY-N	OK	11.78 mm	0.524	0.000
KDS-ST-06-WX-P	OK	8.439 mm	0.311	0.000
KDS-ST-06-WX-N	OK	8.439 mm	0.311	0.000
KDS-ST-06-WY-P	OK	9.727 mm	0.355	0.000
KDS-ST-06-WY-N	OK	9.727 mm	0.355	0.000
KDS-ST-07-EX-P	OK	7.634 mm	0.293	0.000
KDS-ST-07-EX-N	OK	7.634 mm	0.293	0.000

Serviceability Drift Review

Item	Value
Limit	H/200
Status	OK
Max drift	6.948 mm
Max drift ratio	0.00087
Governing	KDS-ST-04-WY-P / story 1

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	8.000 m	0.053 mm	0.00001	0.001	OK
KDS-ST-02	1	8.000 m	0.062 mm	0.00001	0.002	OK
KDS-ST-04-WX-P	1	8.000 m	5.778 mm	0.00072	0.144	OK
KDS-ST-04-WX-N	1	8.000 m	5.778 mm	0.00072	0.144	OK
KDS-ST-04-WY-P	1	8.000 m	6.948 mm	0.00087	0.174	OK
KDS-ST-04-WY-N	1	8.000 m	6.948 mm	0.00087	0.174	OK
KDS-ST-05-EX-P	1	8.000 m	4.517 mm	0.00056	0.113	OK
KDS-ST-05-EX-N	1	8.000 m	4.517 mm	0.00056	0.113	OK
KDS-ST-05-EY-P	1	8.000 m	5.244 mm	0.00066	0.131	OK
KDS-ST-05-EY-N	1	8.000 m	5.244 mm	0.00066	0.131	OK
KDS-ST-06-WX-P	1	8.000 m	5.768 mm	0.00072	0.144	OK
KDS-ST-06-WX-N	1	8.000 m	5.768 mm	0.00072	0.144	OK
KDS-ST-06-WY-P	1	8.000 m	6.929 mm	0.00087	0.173	OK
KDS-ST-06-WY-N	1	8.000 m	6.929 mm	0.00087	0.173	OK
KDS-ST-07-EX-P	1	8.000 m	4.507 mm	0.00056	0.113	OK
KDS-ST-07-EX-N	1	8.000 m	4.507 mm	0.00056	0.113	OK
KDS-ST-07-EY-P	1	8.000 m	5.225 mm	0.00065	0.131	OK
KDS-ST-07-EY-N	1	8.000 m	5.225 mm	0.00065	0.131	OK
KDS-SVC-01	1	8.000 m	0.038 mm	0.00000	0.001	OK
KDS-SVC-02	1	8.000 m	0.048 mm	0.00001	0.001	OK

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M30	OK	0.539	steel-interaction	KDS-ST-04-WY-N
2	M28	OK	0.539	steel-interaction	KDS-ST-04-WY-N
3	M32	OK	0.539	steel-interaction	KDS-ST-04-WY-N
4	M26	OK	0.539	steel-interaction	KDS-ST-04-WY-N
5	M33	OK	0.539	steel-interaction	KDS-ST-04-WY-N
6	M17	OK	0.539	steel-interaction	KDS-ST-04-WY-P
7	M19	OK	0.539	steel-interaction	KDS-ST-04-WY-P
8	M23	OK	0.539	steel-interaction	KDS-ST-04-WY-P
9	M21	OK	0.539	steel-interaction	KDS-ST-04-WY-P
10	M24	OK	0.539	steel-interaction	KDS-ST-04-WY-P

11	M1	OK	0.525	steel-interaction	KDS-ST-04-WX-N
12	M5	OK	0.525	steel-interaction	KDS-ST-04-WX-P
13	M11	OK	0.525	steel-interaction	KDS-ST-04-WX-N
14	M15	OK	0.525	steel-interaction	KDS-ST-04-WX-P
15	M2	OK	0.510	steel-interaction	KDS-ST-04-WX-P
16	M4	OK	0.510	steel-interaction	KDS-ST-04-WX-N
17	M12	OK	0.510	steel-interaction	KDS-ST-04-WX-P
18	M14	OK	0.510	steel-interaction	KDS-ST-04-WX-N
19	M3	OK	0.496	steel-interaction	KDS-ST-04-WX-N
20	M13	OK	0.496	steel-interaction	KDS-ST-04-WX-P

Member Design Trace Matrix

Item	Value
Checked members	37
Unimplemented members	0
NG / WARN	0/0
Max utilization	0.539

Member	Type	Status	Util.	Formula	Action
M30	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M28	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M32	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M26	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M33	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M17	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M19	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M23	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M21	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M24	steel	OK	0.539	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

M1	steel	OK	0.525	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M5	steel	OK	0.525	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M11	steel	OK	0.525	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M15	steel	OK	0.525	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M2	steel	OK	0.510	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M4	steel	OK	0.510	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M12	steel	OK	0.510	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M14	steel	OK	0.510	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M3	steel	OK	0.496	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M13	steel	OK	0.496	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	37	0	0.539
Connections	37	11	1.527
Foundations	15	11	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.

- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.